

Task Grouping Optimization for Gas Insulated Substations Predictive

Maintenance: Trade-off between cost and environmental impact

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Abstract: An effective predictive maintenance plan ensures operational reliability while achieving key optimization goals. In the context of Gas-Insulated Substations (GIS), our core concerns lie in minimizing maintenance costs while mitigating the environmental impact of sulfur hexafluoride (SF_6) gas leakage, a potent greenhouse gas. Building on our previous work optimizing maintenance schedules for individual GIS components, this study extends the approach to a second-stage task-grouping optimization consolidating maintenance tasks across multiple GIS components to address operational inefficiencies such as increased costs, underutilized resources, and prolonged downtime. Specifically, our approach combines maintenance tasks that are temporally close and operationally compatible, thereby reducing travel costs and improving resource utilization. The proposed method formulates the grouping problem as an Integer Linear Programming model, incorporating key constraints such as task time windows, resource availability, and operational priorities. Furthermore, the trade-off between maintenance costs and SF_6 amount leaked is analyzed to account for its environmental impact.

1. Introduction

In modern power systems, Gas-Insulated Substations (GIS) play a critical role in ensuring the reliability and efficiency of electricity transmission and distribution. A GIS is a multi-component system which comprises many Pressurized Metal-Enclosed Compartments (PMEC), which house SF_6 gas-insulated equipment (CIGRE, 2014). As these systems age and their operational complexity increases, effective maintenance planning for PMECs—particularly gas refilling (GR) interventions—is essential to ensure system reliability (Burns, 2024). This study adopts an optimization framework for coordinating maintenance tasks across multiple components, similar to the scalable bi-objective maintenance models proposed by (Kivanç, 2025).

Currently, maintenance tasks have been planned independently for each PMEC based on predefined schedules or reactive responses to alarms. The scheduling of maintenance activities is made possible by the integration of real-time monitoring systems with GIS, as the presence of sensors enables continuous tracking of pressure parameters within each PMEC. Utilizing these data, power utilities have developed alarm prediction algorithms and, based on these forecasts, have established long-term proactive maintenance plans for each PMEC. However, strictly adhering to individual maintenance schedules and addressing each intervention separately is not a globally optimal strategy, as it often results in redundant resource allocation, thus increased operational costs (including transportation, labor, and equipment mobilization), and potential disruptions to power system operations. For instance, dispatching a crew to address a single PMEC's GR incurs certain costs, and recurring visits for temporally proximate tasks increase the total costs. To address this challenge, dynamic update of maintenance planning that groups tasks across multiple PMECs emerges as a strategy to minimize maintenance costs while reducing the frequency of interventions.

This work extends our previous research on predictive maintenance optimization for GIS under operational and environmental constraints. In that study, we developed a framework that provides a decision-supporting Mixed Integer Linear Programming (MILP) optimization model for maintenance scheduling (Wu, et al., 2025). In this paper, we introduced a novel sliding window framework to dynamically group GR tasks for multiple

PMECs. The method identifies tasks scheduled in close temporal proximity, clusters them into intervention batches, and updates subsequent schedules iteratively. By optimizing task grouping within the sliding window, the approach minimizes redundant site visits, thereby reducing transportation costs and labor costs associated with frequent interventions. The sliding window approach proposed in this paper addresses this gap by introducing a flexible and scalable framework for multi-unit system task grouping.

This paper makes the following contributions: (i) The problem of multi-unit maintenance task grouping is formulated as an Integer Linear Programming (ILP) optimization problem with temporal and operational constraints. (ii) A sliding window framework is developed to dynamically identify and group tasks based on their temporal proximity. (iii) The proposed approach is evaluated in the context of GIS maintenance planning using real-world-inspired scenarios, demonstrating its effectiveness in improving efficiency and reducing costs.

2. Problem description

Each PMEC houses electrical components insulated with SF₆ gas, which is a potent greenhouse gas (IPCC, 2018), maintained at pressures between 3.0 and 3.5 bar. While SF₆ is essential for reliable operation, leaks are inevitable due to design and operational factors, posing environmental risks and increasing maintenance demands (Phillip Widger, 2018). Maintenance tasks for PMECS fall into two categories: Gas Refilling (GR): A routine, low-cost intervention triggered when pressure drops to 3.2 bar and typically takes only a few hours to carry out. Repair (RE): A comprehensive repair required after several GR interventions to reduce leakage rates or restore PMECS to near-new condition. RE operations are costly and require extensive planning across multiple departments. Given the impracticality of combining RE operations due to their complexity and long execution times, this study focuses exclusively on optimizing GR task grouping.

The operational challenge arises from the scale and distribution of GIS networks, which often lead to fragmented GR schedules across large numbers of PMECS at different substations. Reactive maintenance practices frequently lead to the mobilization of teams for isolated GR tasks even when multiple maintenance needs occur at the same location within a short time frame. This inefficiency increases transportation costs and personnel deployment frequency while failing to capitalize on opportunities for strategic clustering. By strategically rescheduling GR tasks within acceptable temporal ranges, multiple interventions can be consolidated into single maintenance visits—reducing operational costs without compromising system reliability or safety thresholds. In the present study, we focus on the optimization of task grouping within each site, while the allocation of personnel is regarded as a separate planning process.

3. Methodology

3.1 Sliding window framework

In this section, we present the methodology used to optimize maintenance task grouping across one GIS where all the PMECS are supposed to be in the same location. The sliding window approach serves as the backbone of the proposed methodology, enabling dynamic task grouping over time. The framework divides the entire planning horizon into overlapping time intervals, referred to as "windows." Within each window, maintenance tasks of different PMECS are considered for optimization. Figure 1 illustrates the algorithmic workflow designed to systematically decompose the maintenance grouping problem into manageable temporal segments. It should be noted that the sliding period of the window in this context is flexible, rather than fixed at each iteration. Two scenarios are considered: if there remain ungrouped tasks within the current window, the window slides forward to the position of the next such task; if no such tasks exist, the window advances by the full length of the window. This sliding mechanism ensures that the potential for grouping every task is considered.

The proposed sliding window approach decomposes the global maintenance clustering problem into a series of subproblems. This decomposition stems from two important observations: (i) it is computationally difficult to optimize all maintenance tasks simultaneously in real-world problems due to the long-time intervals (usually measured in months) between actual GR interventions, which makes it impossible to have a clear understanding of the future for a long period of time. (ii) each clustering decision impacts the timing of future tasks. The algorithm maintains the continuity of the solution through a window switching mechanism.

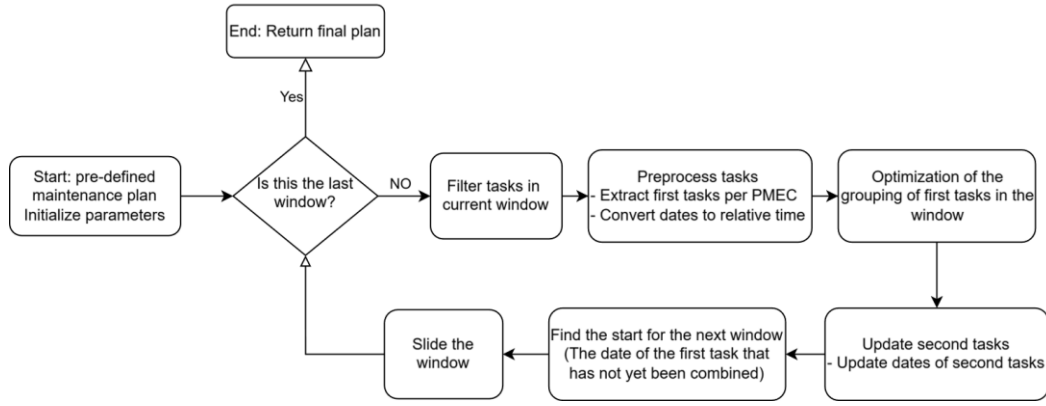


Figure 1. Task grouping in sliding window optimization framework

3.2 Grouping ILP optimization model

In this section, we formulate the grouping optimization problem of one GIS (all PMECs in the same location) as an ILP model that minimizes the number of clusters, i.e., groups as many tasks as possible together while respecting the operational constraints in each window. Table 1 gives the definitions of the parameters and variables.

Problem sets	
N	Set of all the maintenance tasks.
P	Set of PMECs.
Parameters	
T	Planning horizon. The sampling time is set to 1 day.
A	Maximum number of days each task can be advanced.
t_n	Original planned time of task <i>n</i> before grouping.
α_{i(n)}	Degradation rate of PMEC <i>i</i> associated with task <i>n</i> .
C_{max}^{GR}	Maximum number of GR in one cluster.
L	Length of the sliding window.
Integer variables	
a_n	Earliest time that task <i>n</i> can be brought forward to.
b_n	The latest time that task <i>n</i> can be delayed to.
Binary variables	
θ_{nt}	= 1 if a GR task <i>n</i> is assigned to a cluster at time <i>t</i> . Otherwise, 0.
cluster_t	= 1 if cluster at time <i>t</i> is activated. Otherwise, 0.

Table 1. Notations and definitions

The model treats each day in the planning horizon as a potential cluster, i.e., we assume that only one cluster is possible for each day. It is then possible to adopt a time-cluster equivalence principle and unify temporal and cluster indices: a cluster at day *t* is activated if any task *n* is assigned to it (θ_{nt}). This is also the decision variable in the objective function which directly links task-to-cluster.

The proposed ILP model is formulated as follows: the objective function (1) minimizes the total number of activated maintenance clusters across the sliding window, promoting resource consolidation. Constraint (2) enforces cluster activation when containing at least one task, eliminating empty clusters. Each maintenance task receives a unique assignment through constraint (3) to prevent task duplication or omission. The cluster capacity constraint is addressed through constraint (4), which limits the number of tasks assigned to each cluster (i.e., each day), a key operational parameter that reflects resource availability. Constraints (5) to (8) define the time boundaries within which each task can be brought forward or delayed, denoted respectively by *a_n* and *b_n*. Specifically, *a_n* represents the earliest time a task can be brought forward within the scheduling

window. Here, we use the parameter A representing a maximum allowable advancement of 3 days for any task. This value is selected based on the operational needs and restrictions. Considering the length of the window, (5) and (6) effectively define $a_n = \max(0, t_n - A)$, where t_n is the initially scheduled time of task n . Similarly, b_n indicates the latest time task n can be postponed. This value depends on the time remaining before the pressure reaches the critical low-pressure alarm threshold of 3.0 bar. To ensure system safety and avoid reaching this threshold, a conservative safety margin of 2 days is introduced. Constraints (7) and (8) yield b_n as the minimum between this value and the window length L . Temporal feasibility is governed by constraints (9) and (10): if certain tasks are grouped into a cluster scheduled for day t , the day t must fall within the dynamic interval $[a_n, b_n]$ for each task. Otherwise, the constraints are automatically satisfied.

$$\min Obj = \sum_{t=1}^{L+1} cluster_t \quad (1)$$

$$\theta_{nt} \leq cluster_t \quad \forall n \in N \quad \forall t \in [1, L+1] \quad (2)$$

$$\sum_{t=1}^{L+1} \theta_{nt} = 1 \quad \forall n \in N \quad (3)$$

$$\sum_{t=1}^{|N|} \theta_{nt} \leq C_{max}^{GR} \quad \forall t \in [1, L+1] \quad (4)$$

$$a_n \geq t_n - A \quad \forall n \in N \quad (5)$$

$$a_n \geq 0 \quad \forall n \in N \quad (6)$$

$$b_n \leq t_n + (3.2 - 3.0)/\alpha_{i(n)} - 2 \quad \forall n \in N \quad (7)$$

$$b_n \leq L \quad \forall n \in N \quad (8)$$

$$t \theta_{nt} + a_n (1 - \theta_{nt}) \geq a_n \quad \forall n \in N \quad \forall t \in [1, L+1] \quad (9)$$

$$t \theta_{nt} + b_n (1 - \theta_{nt}) \leq b_n \quad \forall n \in N \quad \forall t \in [1, L+1] \quad (10)$$

4. Experiments

4.1 Maintenance plan generation

The maintenance schedule generation simulates degradation patterns and maintenance events for GIS, serving as input data for subsequent optimization. Each P MEC is initialized with randomly generated parameters (drawn from uniform distributions): initial pressure ((3.2–3.5] bar), daily degradation rate ([0.01–0.05] bar/day), and a number between 3 and 5 for the allowable GRs before RE is required. Pressure decreases daily under normal operation until reaching a critical threshold of 3.2 bar, triggering a GR intervention. GR restores pressure linearly to 3.5 bar within one day. Maintenance events are recorded in the form of a data frame. The parameter values used in this study (shown in Table 2) are hypothetical and designed to suit the small-scale case. However, they are based on experience drawn from similar real-world situations. These values were carefully chosen to match the experiment size and to clearly demonstrate the model's functionality in a simplified yet representative way.

Parameter	Value (unit)	Parameter	Value (unit)
C_{max}^{CR}	5 (task)	T	30 (day)
L	10 (day)	$\alpha_{i(n)}$	[0.01, 0.05] (bar/day)
A	3 (day)	$ P $	10 (unit)
$ N $	22 (task)		

Table 2. Table of parameters with units

4.2 Grouping results

Figure 2 illustrates the progressive clustering of maintenance tasks across the planning horizon, with each subfigure representing a successive window execution. The first panel of Figure 2 presents the initial

distribution of tasks across 10 PMECs, showing a temporally dispersed pattern that would traditionally necessitate 22 separate maintenance deployments. This fragmentation represents the baseline scenario where each task would be executed at its originally scheduled time without any strategic grouping. The subsequent panels reveal the sequential operation of the sliding window, with each red-bordered rectangle representing the active window under consideration. The green circles indicate the positions to which the tasks have been adjusted through the clustering process, with the arrows visualizing the direction of these adjustments. The algorithm systematically processes consecutive windows, identifying possible grouping opportunities while respecting the maximum allowable time-shift constraints.

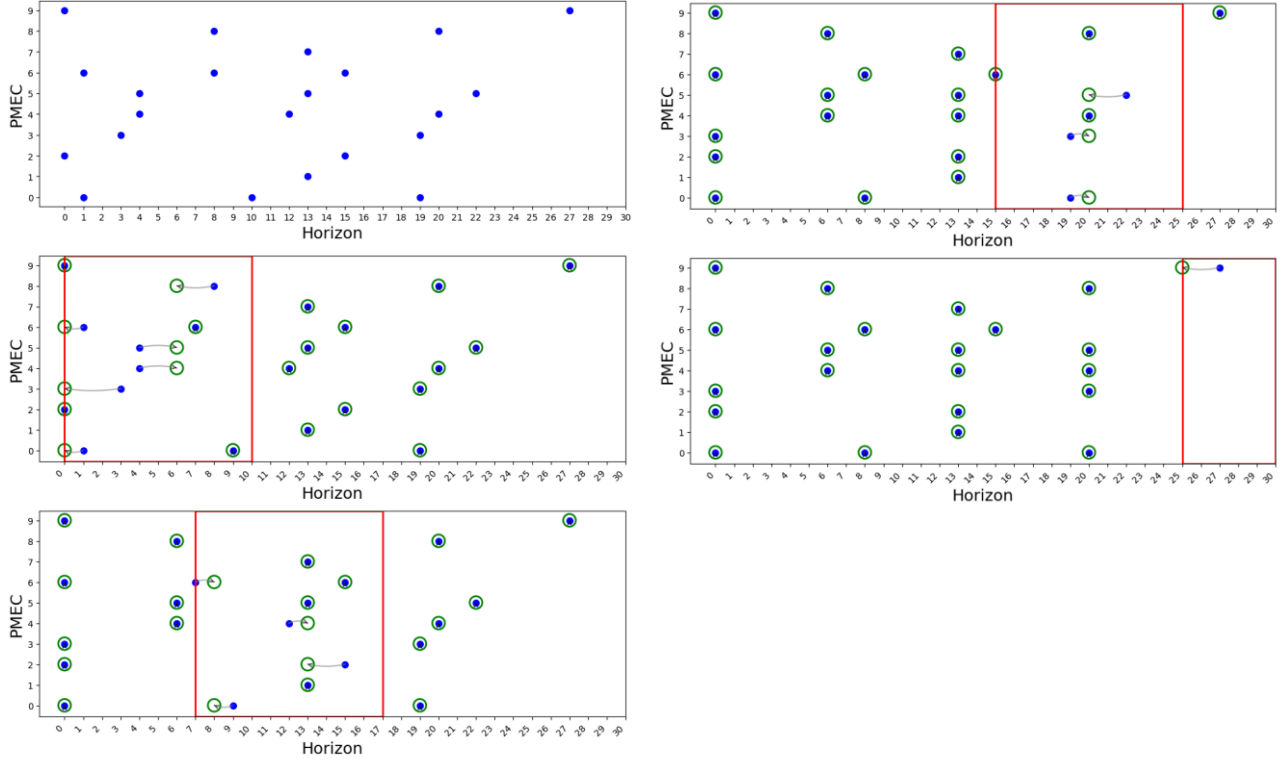


Figure 2. Sliding window grouping results

Figure 3 illustrates the temporal adjustments during clustering, comparing original task schedules to their assigned cluster centers. The algorithm optimizes task grouping by advancing or postponing tasks within a restrained range, adhering to operational constraints. A slight preference for advancing tasks is observed, reflecting a priority for prompt maintenance to ensure system reliability.

Economically, the clustering approach achieves significant cost savings. By consolidating 22 individual GR tasks (each costing 10 units) into 7 maintenance clusters, total expenditure is reduced from 220 units to 70 units—a 68.2% reduction. This efficiency stems from eliminating redundant transportation and setup operations while keeping interventions within acceptable time frames. These results validate the method's practicality for reducing costs in large-scale GIS networks without compromising reliability. The cost reduction discussed here primarily refers to costs associated with transportation and personnel, while the execution cost of the tasks themselves remains unaffected. It should also be noted that the potential increase in costs resulting from a greater number of tasks being executed on the same day due to grouping is currently beyond the scope of our study.

Environmentally, SF₆ gas leakage increased slightly from 42.114 kg to 42.696 kg post-clustering (1.38%). Our analysis across multiple scenarios indicates this is not a consistent trend, suggesting the environmental impact of maintenance clustering is context-dependent rather than systematic. The observed variation likely stems from two key limitations in our current modeling approach: (i) constant leakage rate assumption: our model presumes fixed leakage rates for each PMEC throughout the planning horizon, which may not accurately reflect dynamic degradation patterns in real-world systems. (ii) maintenance interdependency: the current formulation does not account for potential interactions between maintenance types, particularly how the timing of GR operations might influence the scheduling of RE activities, which directly affect leakage rates.

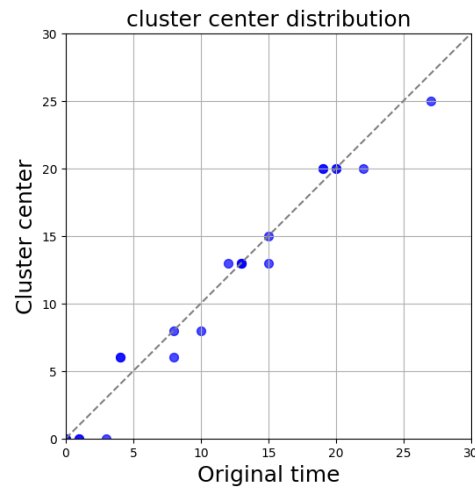


Figure 3. Cluster center vs. original time

5. Conclusion and discussion

This study demonstrates that clustering GIS maintenance tasks using a sliding window algorithm significantly improves operational efficiency, reducing deployment costs by 68.2% while maintaining system reliability. The method consolidates temporally dispersed GR tasks into coordinated interventions, minimizing redundant site visits. Although minor variations in SF₆ leakage were observed post-clustering, these were context-dependent, highlighting the need for dynamic leakage models in future research. The framework offers a scalable solution for optimizing maintenance logistics in large-scale power networks, balancing economic and operational constraints.

The current solution framework may yield multiple equivalent solutions, as it lacks explicit preferences for advancing or delaying maintenance tasks. In practice, certain scheduling behaviors may be preferred, so future work could introduce behavior-specific penalties or preferences to better align with operational priorities. The model's daily maximum capacity can also lead to uneven cluster distributions, with some clusters overloaded and others containing only isolated tasks. Incorporating soft balancing constraints or fairness considerations could improve efficiency and practicality. Furthermore, the framework does not yet account for logistical constraints such as the geographic dispersion of substations and associated travel times. Factoring in these real-world logistical considerations would make the model more realistic and enhance its applicability to large-scale GIS maintenance planning.

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